HARSHIT – AI ORCHESTRATOR (Brain + Logic)

you will build the core intelligence layer that converts human prompts into structured scientific outputs. Using advanced language models, you will design a system that understands user requirements and transforms them into precise JSON specifications. You will integrate workflow management to handle retries and logic flow, connect with vector databases to fetch the most relevant traits, and use research APIs to enrich outputs with the latest scientific knowledge. Your focus is to ensure that every input is processed intelligently and produces a clean, reliable spec.json in seconds, forming the foundation for the simulation system.

🎯 **Your Goal:**

Convert **human prompts → structured scientific JSON (spec.json)**
You are the **main brain that understands and thinks**

✅ **Tasks to Complete**

🔹 **1. Prompt Understanding System (LLM)**

- Use:
  - DeepSeek-R1 / Llama 3.5
- Convert user input into structured JSON
- Example:
  - Input: "wheat for Jodhpur at 48°C"
  - Output: structured parameters

🔹 **2. Workflow System (LangGraph)**

- Build agent flow:

Input → Process → Check → Retry (if failed)

- If output is bad → loop back and fix
- Automate retries and corrections

🔹 **3. Vector Search System**

- Use:
  - ChromaDB + Sentence Transformers
- Convert traits into embeddings
- Find similar traits from 1M dataset in milliseconds

🔹 **4. Research Integration**

- Use:
  - Semantic Scholar API / ArXiv
- Fetch:
  - latest scientific insights
- Extract only useful information (not full papers)

🔹 **5. JSON Output System**

- Use:
  - Pydantic
- Force strict output format:

spec.json

- No free text allowed

🔹 **6. Final Spec Generator**

- Combine:
  - prompt understanding
  - vector results
  - research insights
- Output:
  - final validated spec.json

🔹 **7. Testing Interface**

- Use:
  - Streamlit
- Create simple UI:
  - input → output

🔗 **Dependency**

- Takes data insights from **Jay**
- Sends final spec → to simulation team (Srikar side)

⚙️ **JAY – DATA + SEARCH ENGINE (Support Brain)**

you will build the data backbone that powers fast and accurate retrieval of information. You will convert large-scale trait data into vector embeddings and store them in a high-performance vector database for instant similarity search. Your system will allow Harshit to find the most relevant traits from millions of records within milliseconds. You will also handle data cleaning, structuring, and optimization to ensure smooth integration across the pipeline. Your goal is to make the entire system efficient, scalable, and reliable by providing the right data at the right time.

🎯 **Your Goal:**

Provide **fast, accurate data + similarity search + knowledge support**

✅ **Tasks to Complete**

🔹 **1. Trait Embedding System**

- Use:
  - Sentence Transformers
- Convert all traits into vectors
- Store embeddings properly

🔹 **2. Vector Database Setup**

- Use:
  - ChromaDB (or Qdrant if needed)
- Store:
  - 1M traits embeddings
- Enable fast similarity search

🔹 **3. Similarity Search API**

- Build function:

input: condition (e.g. Jodhpur heat)

output: top similar traits

- Target:

    - <50ms search time

🔹 **4. Data Cleaning + Structuring**

- Clean Kartik's Parquet data
- Ensure:
    - correct fields
    - no missing values
    - consistent format

🔹 **5. Research Data Processing**

- Help Harshit by:
    - filtering research results
    - extracting key points
- Provide structured summaries

🔹 **6. Integration Layer**

- Connect:
    - vector DB → Harshit's system
- Ensure:
    - smooth API/data flow

🔹 **7. Performance Optimization**

- Ensure:
    - fast queries
    - scalable system
- Handle large dataset efficiently

🔥 **FINAL FLOW (VERY IMPORTANT)**

User Input

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Harshit (LLM + Logic)

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Jay (Search + Data)

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Harshit (Final JSON Output)

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Simulation Team (Srikar system)

🎯 **FINAL OUTPUT**

- ✅ Prompt → JSON system working
- ✅ Vector search on 1M traits
- ✅ Research integration
- ✅ Clean structured output
- ✅ End-to-end pipeline